AGENTX HACKATHON 2026 · PS-04 SUBMISSION

ReturnLens: Evidence-Verifiable Return & Refund Agent

Verifying customer video evidence against mock store truth before any refund or label is issued.

Problem Statement

PS-04 · Autonomous support agent for online sellers
Narrowed Wedge: Evidence-verified return & refund triage.

Team Information

Team: Team Rookie
Members: Nam Khánh Lê & Khiêm
Institution: University of Science (HCMUS)

Core Architecture

Fail-Closed Verification: Deterministic policy engine + timestamped video grounding + 1-click human gate.
Zero GPU required.

01 · Problem Fit & Real-World Impact

The High-Cost Blindspot of E-Commerce Returns

The Solo Seller Bottleneck & Double-Sided Loss

- **Overwhelmed Solo Sellers:**
Sellers managing 50–500 orders/month handle returns alone: watching 30–60s unboxing clips, cross-referencing order tabs, and calculating policy dates manually.
- **Double-Sided Financial Risk:**
Wrongly approving fraudulent returns ('wardrobing', worn items, swapped goods, stripped tags) destroys profit margins; wrongly rejecting buyers triggers marketplace penalties & disputes.
- **Lack of Dispute Evidence:**
UniCommerce reports that sellers routinely lose return claims due to 'lack of concrete evidence' to contest invalid claims.
- **What Is NOT the Problem:**
Not another WISMO ('Where Is My Order') chatbot. Commercial bots (Robylon, Robofy) already handle basic tracking. The real unresolved pain is verifying return claims.

The ReturnLens Narrowed Solution

✓ **Evidence-Verifiable Triage:**
ReturnLens inspects customer unboxing videos, cross-checks against actual order data, and validates claims before any action is suggested.

✓ **Fail-Closed Safety Guarantee:**
If evidence is ambiguous or contradictory, the agent escalates to the seller with timestamped proof clips. It NEVER hallucinates a rejection or issues an unverified refund.

✓ **10-Second Decision Package:**
Compresses a 5-minute manual review into a 10-second 1-click decision for the seller, complete with exact video timestamps and policy citations.

✓ **Fairness to Buyers as a Constraint:**
Escalation ensures legitimate buyers with blurry clips are asked for specific photos rather than being blindly rejected by an algorithm.

02 · Solution Architecture & End-to-End Workflow

5-Step Verifiable Triage: From Customer Video to 1-Click Action

01

Step 1: Intake

Customer submits return claim via web chat:
• Reason (damaged/wrong item)
• 30–60s video or photo
• Order ID reference

02

Step 2: Typed Plan

Planner emits structured plan object:
• Sets refund ceiling
• Specifies required checks
• Defines tool calling path

03

Step 3: Grounding

Retrieves dual truth:
• Mock Store API (SKU, price, delivery date)
• CPU ffmpeg frame extraction (~1 fps) + multimodal LLM

04

Step 4: Verify

Deterministic Policy Engine:
• Verifies return window (≤ 7d)
• Verifies refund amount ≤ cap
• Matches SKU & damage timestamps

05

Step 5: Action

Human Approval Gate:
• Within policy → 1-click approve → mock label & refund
• Discrepancy → Escalate with timestamped proof pack

03 · Agent Design: Planner → Tools → Verifier → Human Gate

Deterministic Policy Verification & The Fail-Closed Principle

A. Planner (Typed JSON)

- Generates structured typed plan (steps, expected outputs, refund caps)
- Enforces policy constraints upfront
- Code never evaluates freeform prose

B. Tool Execution Layer

- Mock Store & Shipping API: FastAPI + SQLite seed data (orders, delivery dates)
- Video Indexer: CPU ffmpeg frame extraction + free-tier multimodal LLM
- Audit Logger: Immutable JSONL trail

C. Deterministic Verifier (Code)

- Pure code — zero LLM override:
  1. Order exists & matches buyer
  2. Within return window (≤ 7 days)
  3. Refund amount ≤ cap & order total
  4. Every claim backed by tool output

D. Critic & Human Approval Gate

- Independent Critic LLM (temp 0) audits for claim-evidence mismatches
- Human-in-the-Loop interrupt before any money or label is issued
- Turns review into a 10s 1-click decision

THE FAIL-CLOSED PRINCIPLE: "If evidence, policy, or checkers disagree — we escalate. Never auto-reject a buyer; never auto-refund on weak evidence."

04 · Evidence Proof Package & 3 Demo Scenarios

Timestamped Video Grounding & Robust Fallback Handling

Case 1 · Clear Approval (45s)

Buyer reports torn seam with 35s unboxing video:
• Order verified delivered 3 days ago (within 7d window)
• Damage detected at 0:12–0:18 with high confidence
• Refund within category ceiling (₹850 ≤ ₹1,500)
• Seller 1-click approves → mock return label issued & logged.

Case 2 · Star Case: Escalated (60s)

Buyer claims wrong item; video shows mismatched SKU & missing original brand tag:
• Policy requires tags attached for apparel returns
• Deterministic verifier rejects auto-approval
• Agent ESCALATES to seller with proof pack & timestamp citations (does NOT auto-reject buyer).

Case 3 · Missing Evidence (45s)

Video too dark / blurry to discern item condition:
• Verifier identifies confidence < threshold
• Agent asks buyer for specific photo of brand tag
• Case remains open with checklist
• Closing 30s: Live audit trail & seller time savings.

Mock Data Integrity & Fallback Matrix

• Mock Store & Videos: Staged demo video clips & SQLite mock database (no real customer PII, 100% compliant with hackathon rules).
• Graceful Fallback: Multimodal model timeout → degrade confidence & escalate to seller; shipping API unconfirmed → escalate. Zero blind guesses.

05 · 24-Hour Sprint Plan & PS-04 Compliance

Realistic 2-Person Division of Labor & Zero-GPU Architecture

24-Hour Build Sprint Plan (11-12 Nov)

⏱ **Hours 00 – 06: Core StateGraph & Mock Store API**

Member A builds LangGraph state machine & SQLite/FastAPI store. Member B builds CPU ffmpeg video frame extractor.

⏱ **Hours 06 – 12: Grounding & Deterministic Verifier**

Member A implements hardcoded policy engine (caps, dates). Member B integrates Gemini/Groq free tier multimodal grounding.

⏱ **Hours 12 – 18: Dashboard & Human Approval UI**

Member A wires audit logging & Critic agent. Member B builds Streamlit/React seller dashboard with timestamped proof clips.

⏱ **Hours 18 – 22: End-to-End Testing & 3 Demo Cases**

Dry run across the 3 benchmark cases (Approved, Escalated, Incomplete). Validate 100% test pass rate.

⏱ **Hours 22 – 24: Demo Recording & Final Packaging**

Record 3-minute video walk-through, polish GitHub README, verify submission links.

PS-04 Must-Have Checklist & Stack

MANDATORY REQUIREMENTS:

✓ **Tool Calls on Mock Store:**
Verified via FastAPI + SQLite seed

✓ **Refund-Limit Guardrails:**
Hardcoded in deterministic code

✓ **Log Every Action Taken:**
Immutable append-only JSONL audit log

✓ **Human Confirmation Step:**
Mandatory 1-click seller sign-off

✓ **Zero GPU Dependency:**
CPU ffmpeg + free tier hosted LLMs

TECH STACK: Python 3.12, LangGraph, FastAPI, SQLite, ffmpeg, Gemini 2.5 Flash, Streamlit.